Zen Nguyen

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TECHNICAL SKILLS

- **Languages:** Python, Go, Java, C++, TypeScript, JavaScript, SQL
- **AI & Data:** RAG Architecture, Vector Databases, LangChain, OpenAI API
- **Backend & Systems:** FastAPI, Node.js, Express, RESTful APIs, gRPC, TCP Sockets, Raft Consensus Algorithm
- **Infrastructure & DB:** AWS, Docker, PostgreSQL, MongoDB, Git, Linux
- **Testing & Validation:** JUnit, Pytest

TECHNICAL PROJECTS

TitanSwarm (Autonomous AI Co-Pilot) & TitanStore (Raft Database) Jan 2026 – Present

Personal Project · Python, Go, LangChain, FAISS

- Automated thousands of Software Engineering job postings by architecting an autonomous AI Co-Pilot in Python, leveraging **CI/CD pipelines** via concurrent JobSpy scrapers for data acquisition.
- Delivered uniquely tailored, ATS-optimized PDF resumes by building a zero-hallucination RAG engine using LangChain, OpenAI APIs, and FAISS, ensuring data integrity.
- Ensured application state durability and fault tolerance across network failures by developing TitanStore, a distributed memory bank in Go, implementing the **Raft Consensus Algorithm** and a custom Write-Ahead Log (WAL).
- Enabled concurrent, real-time status updates to the Go database cluster by designing and implementing a high-throughput TCP client API, supporting seamless scaling.

SFU Course Tracker (sfucourseplanner.me) Nov 2025 – Jan 2026

Personal Project · Python, Docker, AWS, FastAPI

- Optimized deployments and reduced infrastructure costs by migrating a full-stack platform from Azure to **AWS (EC2)**, implementing **Docker** container orchestration.
- Increased data throughput by 10x while respecting server constraints by engineering a scraping pipeline using Asyncio and HTTPX with semaphore-based rate limiting.
- Converted unstructured text into a deterministic Abstract Syntax Tree (AST) by designing and implementing a custom parser to tokenize and evaluate nested boolean prerequisite strings.
- Optimized read performance for complex queries by designing a normalized schema using SQLModel with JSON-type columns to store recursive tree structures.

Gridlock Casino (2D Arcade Engine) Feb 2026 – Present

Collaborative Project · Java, Swing, Maven, JUnit

- Managed project build automation and dependency resolution using Apache Maven to synchronize a 6-developer team, ensuring adherence to **CI/CD methodologies**.
- Validated core engine logic through comprehensive **JUnit testing**, ensuring reliability and system stability.

EDUCATION

Bachelor of Science, Computing Science May 2025 – Present

Simon Fraser University

- CGPA: 3.74 / 4.33
- Relevant Coursework: Discrete Math (A+), Computer Systems(A), Data Structures & Programming (B+), Linear Algebra(A-).
- Langara College | Associate of Science, Computer Science Jan 2024 – Apr 2025